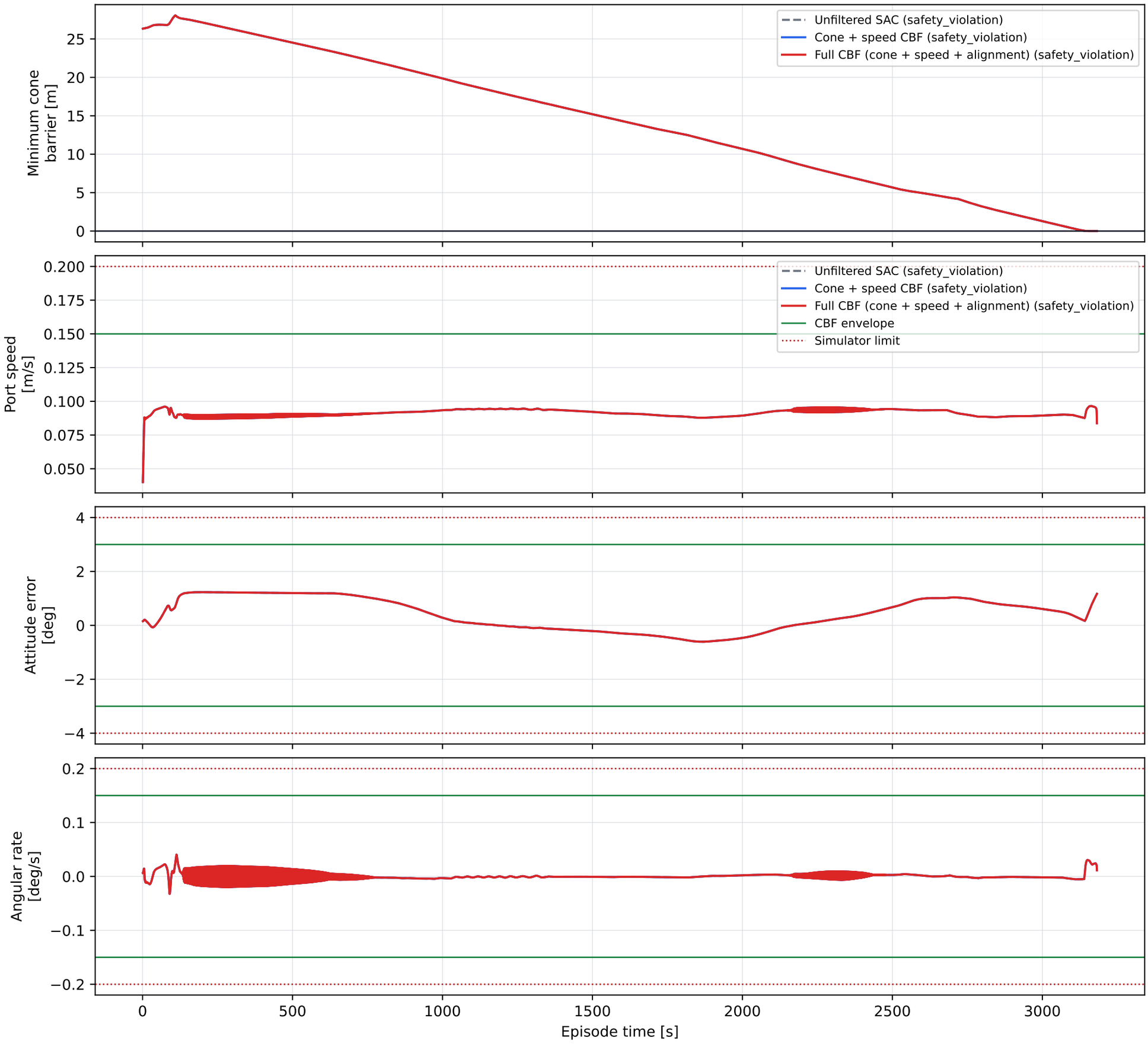

Protected-state envelopes — evaluation seed 120
green: CBF boundary; dotted red: simulator failure boundary
Minimum cone barrier [m]
Unfiltered SAC (safety_violation)
Cone + speed CBF (safety_violation)
Full CBF (cone + speed + alignment) (safety_violation)
Port speed [m/s]
CBF envelope
Simulator limit
Attitude error [deg]
Angular rate [deg/s]
Episode time [s]